

Data Structures and Algorithms

Data structures and algorithms are commonly heard by programmers, whether they code in Java, Python, C, or many other languages. This is because these two are key elements in programming. **Data structures** are used to store and organize data in order to use it properly and make it much more understandable, while **algorithms** are used to execute instructions in a specific order or set of rules (Data Structure and Algorithms Tutorial - Tutorialspoint, n.d.).

There are many different types of data structures. One of them is called **Linear Data Structure**. This is where data is arranged in a sequential order, meaning that elements are placed one after the other. Examples of these are arrays, linked lists, stacks, and queues (Unstop - Competitions, Quizzes, Hackathons, Scholarships and Internships for Students and Corporates, n.d.). One example is the **One-Dimensional Array**. It consists of only a single row of elements starting from index zero and continuing onward. An example of its syntax is “data_type name[size] = {value1, value2, ...};”, and in actual code, “int numbers[5] = {85, 90, 78, 92, 88};” (GeeksforGeeks, 2024).

Apart from data structures, it is also important to showcase the efficiency of your algorithm. The **Big O notation** is a concept for analyzing how efficient an algorithm is. It helps developers understand how time and space complexity change as the input size increases. It's important for programmers because it helps developers analyze the performance of their algorithms and improves the performance of the actual program (Van, 2025).

Moving on to searching algorithms, what are these exactly? Based on the information provided by GeeksforGeeks (2024), **Searching algorithms** are key components used for searching data in a stored data collection. When searching for specific data, there are two commonly used algorithms based on the type of array used. If it is for an unsorted array, linear search will be used, while binary search for sorted arrays. **Linear search** works by comparing each item one by one, until it matches the item being searched. For example, there are four elements in an array. If we search for an element, it will start from the first value up to the value that the user searches for. **Binary search** works by comparing the element in the middle part first and if it matches with what the user searched for, it returns. If not, it divides the search into left and right,

depending on if the middle element is larger or smaller than the result. For example, we have numbers 10, 20, 30, 40, 50, 60, 70, and we search for 60, it will first check 40 (the middle), and since it doesn't match, it will ignore the left part (10 – 30), including 40 since it's smaller than the result. Now that 50, 60, and 70 are left, it will check the middle again, which is now 60. Since it matched, it will return the result. Basically, it keeps cutting the data into half, until it reaches the item being searched for.

To conclude, data structures and algorithms are both important in programming, as they help programmers organize data, create efficient algorithms, and search for specific information. Understanding this can help improve the performance and efficiency of programs being created.

References

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